MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMAT202
OPERATIONS RESEARCH

QP CODE : 25901391

(2024 ADMISSION ONWARDS)

Duration: 2 hrs Maximum Marks: 70

Part A
Answer any FIVE questions. Each question carries 2 marks

1. Infeasibility is a condition that exists when there is no solution to an LP problem that satisfies all the constraints and non-negativity restrictions. 2Mark

2. The coefficients of decision variables in the objective function. 1Marks
   Any correct example 1 Mark

3. An LPP has no limit on the constraints, i.e.,the feasible region is not bounded 2Marks

4. Z_j- row co-efficients under any column are obtained by adding the products of elements under that column with the corresponding C_B value $Z_j = \sum C_B a_{ij}$, where a_{ij} is the matrix element in the ith row and jth column 2Marks

5. The total availability may not be equal to the total demand $\sum_{i=1}^{m} a_i \neq \sum_{j=1}^{n} b_j$ 2Marks

6. Unbalanced $\sum_{i=1}^{m} a_i \neq \sum_{j=1}^{n} b_j$ 1Mark
   balanced $\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$ 1Mark

7. 2Marks

	1	2	3	4
1	2	0	1	4
2	0	1	4	2
3	3	2	0	1
4	1	3	0	2

8. Maximization problem has to be changed to minimization before the Hungarian method may be applied. This transformation may be done in either of the following two ways.
(i) By subtracting all the elements from the largest element of the matrix.
(ii) By multiplying the matrix elements by -1. 2Marks

Part B
Answer any FIVE questions. Each question carries 6 marks
Maximum Marks from the section is 30

9. (i) Any unrestricted variable x_j can be expressed as the difference of two non-negative variables. i.e., $x_j = x_j^{'} - x_j^{"}, x_j^{'}, x_j^{"} \geq 0$
(ii) If R.H.S. of a contraint os negative, it is multiplied on both sides by -1 to make it positive. This will reverse the sign of inequality
(iii) The inequality contraints are changed to equality constraints by adding or subtracting a non -negative variable from the L.H.S of such constraints 4Marks
Any correct example 2Marks

10. Correct graph 4Marks
A(0,0) Z = 0
B(6,0) Z = 12
C(6,4) Z = 28
D(2,8) Z = 36
E(0,8) Z = 32
Max at (2,8) and Max value of Z = 36 2Mark

11. Max $Z = 3x_1 - x_2$ subject to
$2x_1 + x_2 \leq 2$, $x_1 + 3x_2 \leq 1$, $x_2 \leq 4$, $x_1, x_2 \geq 0$
$2x_1 + x_2 + s_1 = 2$, $x_1 + 3x_2 + s_2 = 1$, $0x_1 + x_2 + s_3 = 4$
$s_1 = 2,\ s_2 = 1,\ s_3 = 4$ 3Mark
Problem has non-degenerate basic feasible solution 1Mark

Because in basic feasible solution all the three variables are positive. 2Marks

12. Min $Z = 3x_1 - 2x_2 + x_3^{'} - x_3^{"}$ subject to $-2x_1 + 3x_2 - x_3^{'} + x_3^{"} \geq -5$, $4x_1 - 2x_2 \geq 9$, $-8x_1 + 4x_2 + 3x_3^{'} - 3x_3^{"} \geq 8$ 2Marks
Max $W = -5y_1 + 9y_2 + 8(y_3^{'} - y_3^{"})$ subject to
$-2y_1+4y_2-8(y_3^{'}-y_3^{"}) \leq 3$, $3y_1-2y_2+4(y_3^{'}-y_3^{"}) \leq -2$, $-y_1-3(y_3^{'}-y_3^{"}) \leq -1$, $y_1, y_2, y_3^{'}, y_3^{"} \geq 0$ 2 Marks
Max $Z^* = -5y_1 + 9y_2 + 8y_3$ subject to
$-2y_1 + 4y_2 - 8y_3 \leq 3$, $3y_1 - 2y_2 + 4y_3 \leq -2$, $-y_1 + 3y_3 = 1$, $y_1, y_2 \geq 0$, y_3 unrestricted in sign. 2Marks

13. Allotment using VAM. $a_{12} = 15, a_{13} = 5, a_{21} = 10, a_{13} = 15, a_{14} = 5, a_{34} = 25$ 4Marks
There is no degeneracy. Because there are six alloted cells. 1Mark
To remove the degeneracy put an ϵ in the independent position. 1Mark

14. (a) Cost Matrix 3Marks

	D1	D2	D3
S1	4	8	6
S2	10	2	7
S3	6	4	0

To obtain an initial basic feasible solution 3Marks
(c) For tesing the optimality 3Marks
Maximum profit is 2960. 2Marks[MAX MARKS 6]

15. 3Tables 3 Marks
Job A to Z, Job B to Y, Job C to X 2Marks
Min cost =47 1 Mark

16. Theorem 1 : In an A.P, if we add or subtract a constant to every element of a row(or column) in the cost matrix, thenan assignment which minimizes the total cost on one matrix also minimizes the total cost on the other matrix. 3Marks
Theorem 2 : If all $c_{ij} \geq 0$ and we can find a set $x_{ij} = x_{ij}^*$ such that $\sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}^* 0$ then this solution is optimal. 3Marks

Part C
Answer any THREE questions. Each question carries 10 marks
Maximum Marks from the section is 30

17. Max $Z = 90x_1 + 40x_2 + 30x_3$, subject to $6x_1 + 5x_2 + 2x_3 \leq 5000$, $4x_1 + 7x_2 + 3x_3 \leq 6000$. 5Marks
$x_1 : x_2 : x_3 :: 3 : 4 : 5$ or $\frac{x_1}{3} = \frac{x_2}{4} = \frac{x_3}{5}$
Additional constraints $4x_1 - 3x_2 = 0$, $5x_1 - 4x_3 = 0$, $x_1, x_2, x_3 \geq 0$ 5Marks

18. Basic solution: A solution obtained by setting any n variables(among m+n) equal to zero and solving for remaining m variables 2Marks
Feasible solution: A set of variables satisfies the constraints as well as non negative restrictions 2Marks
Basic feasible solution: A basic solution that also satisfies the non negativity restrictions.Every BFS is an extreme point of the convex set of feasible solution. 2Marks
Non-degenerate BFS : It is a BFS in which all the m basic variables are positive and the remaining n variables are zero each. 2Marks
Degenerate BFS: It is a BFS in which one or more of the m basic variables are equal to zero. 2Marks

19. $s_1 = 2, A_1 = 3, s_3 = 4$ 1Mark
3 Tables 2 Marks each
$x_1 = 3/5$ and $x_2 = 4/5$, $MaxZ = 1$ 3 Marks

20. VAM 5Marks
2Tables of MODI method 2Marks each
cost = 743 1Mark

21. Steps of Hungarian Method
First 3 steps 6Marks
Next 3 steps 4Marks
OR
Draw the correct flow chart for Hungarian method
Correct upto second decision box 5 Marks
Correct upto third decision box 8 Marks
Correct flow chart 10 Marks

22. Explain Meaning of AP 5Marks
Write Mathematical method 5Marks

N.B Suitable stage marks may be fixed by the examiner for alternate correct answers.